Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Matthias Trenn	**Email**	matthias.trenn@ilt.fraunhofer.de
Institute Name	Fraunhofer Institute for Laser Technology ILT			
Address	Steinbachstraße 15, 52074 Aachen, Germany			

2. Technology Overview

Title of Proposed Technology	*High rate laser based battery structuring for power density enlargement in industrial battery production*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Secondary Batteries*	*Batteries*	*Manufacturing and Measurement/Evaluation Equipment*
Concept of Proposed Technology	• *The laser creates a 3D structure in the electrodes in the form of trenches and holes, which enables the ions to occupy the active material more quickly.* • *Very short pulses are used so that the material is selectively vaporized and the surrounding material is not thermally damaged.* • *The advantages of this process have been proven in numerous studies: It increases the charging and discharging rate without capacity loss, reduces the occurrence of defects and thus increases service life, and also reduces formation time.*			

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Rationale for Joint R&D Project	• *Even though positive results have been demonstrated in many studies, industrial implementation has not yet taken place. Currently, the main challenge lies in the available laser power and optical process control of the systems.* • *Fraunhofer ILT is the only institute to have an industrial high-power laser in the ultrashort pulse range with laser power of up to 1 kW. In addition, Fraunhofer ILT has many years of experience and expertise in the development of optical systems for large-area industrial processing and laser-material interaction.* • *Fraunhofer ILT is looking for a partner from the battery industry to develop an industrial system and process using laser systems to enable the industrial production of microstructured electrodes.*
Key Words	1. *Battery improvement* 2. Decrease of formation time 3. Faster charging and decharging 4. Increase in service life 5. High rate laser structuring

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level ( )	End of the Project	Level ( )
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project*			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters*			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives*			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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